

YEAR 7 • UNIT 1 • HOMEWORK 5

Unit 1, All of It

Science 1.1–1.4 — cells, specialised cells, tissues and organs

Mathematics 1.1–1.6 — integers, factors, multiples, tests and roots

Name: _____

Class: _____

Date given: _____

Date due: _____

How to do this homework

This packet is **the whole of Unit 1**, in one place. Science is at the front this time; the maths is longer, so it is at the back.

- In maths, **write the calculation before the answer**. Every time.
- In science, answer in **full English sentences**. “Organ” is not a sentence; “The heart is an organ” is.

Section A • Science 1.1–1.4 — Cells, Tissues and Organs**A1 Match the word to its meaning**Write the correct **letter** in each box.

Word		Meaning
1. Organelle	☐	A A group of organs working together, like the heart and the blood vessels.
2. Tissue	☐	B The strong material a plant cell wall is made of.
3. Organ	☐	C A whole living thing, like a frog or an oak tree.
4. Organ system	☐	D A part inside a cell that does one job.
5. Organism	☐	E Changed in shape so it can do one job very well.
6. Chlorophyll	☐	F A group of similar cells working together, like muscle.
7. Specialised	☐	G The green substance inside a chloroplast that traps light.
8. Cellulose	☐	H Several tissues working together, like the heart or a leaf.

A2 Climb the ladder

Word help – the ladder, in order

cell → **tissue** → **organ** → **organ system** → **organism** – and each rung is **made of** the rung below it. A tissue is made of cells; an organ is made of tissues.

For each thing, write which **level** it is, and then what the **level above it** would be. One of these has no level above it – think carefully before you write.

Thing	Which level?	The level above it
A red blood cell		
Muscle		
The stomach		
The digestive system		
Mr Bowen		

A3 Name the cell

Remember from the lesson

A cell's **function** is the job it does. A **specialised** cell has a structure that helps it do that job well. The sentence to use: A [cell] is **adapted to** [its job] **because it has** [its feature].

(a) Each clue describes one specialised cell. Name it.

(i) It has no nucleus, which leaves more room for the oxygen it carries. _____

(ii) It is very long and thin, so a message travels a long way along it. _____

(iii) It has tiny hairs along the top that sweep dust out of your lungs. _____

(iv) It has a long thin part that reaches into the soil for water. _____

(v) It is packed with chloroplasts and sits near the top of a leaf. _____

(b) Now write the full sentence for **two** of them, using **adapted to** and **because it has**.

(c) A palisade cell is full of chloroplasts. A root hair cell has **none at all**. Suggest why not.

A4 Answer in full sentences**Sentence starters – you may copy these**

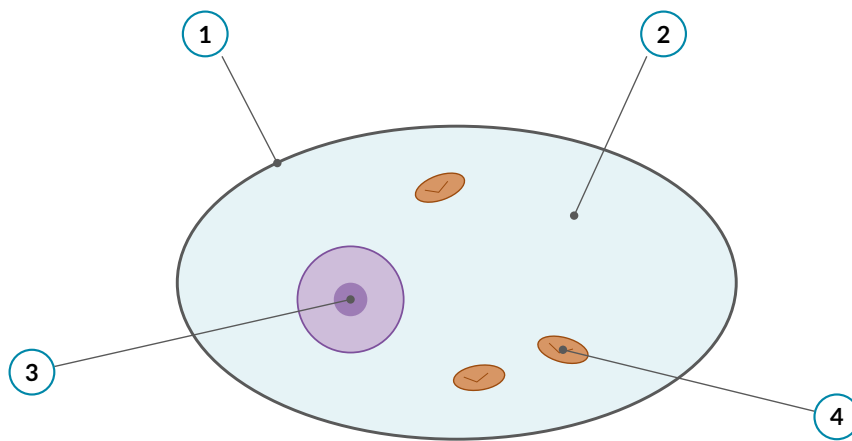
“I know it is a ... cell because...”

“It also has..., and an animal cell has no...”

Mr Bowen looks at one cell under the microscope. It has a straight, stiff edge, and one large space fills most of the middle of it. Is it a **plant** cell or an **animal** cell? Say **how you know**, in full sentences.

A5 Label the animal cell

Write the name of each numbered part on the lines below the diagram.



1. _____

3. _____

2. _____

4. _____

Now think. Name **two** things a **plant** cell would have that this cell does not.

Section B · Mathematics 1.1–1.2 – Integers

The rules are on the next page. Try each question **before** you look at them – if you can, you know them.

The two sets of rules, side by side

Add a positive · move **right** $(+) \times (+) = (+)$

Add a negative · move **left** $(-) \times (-) = (+)$

Subtract a positive · move **left** $(+) \times (-) = (-)$

Subtract a negative · move **right** $(-) \times (+) = (-)$

Two negatives make a positive is true for $\times$ and $\div$. It is **not** true for $+$: $-6 + (-4) = -10$, not -2 .

B1 Higher, lower, warmer, colder

Complete each sentence with **higher than**, **lower than**, **warmer than** or **colder than**.

(i) -9 is _____ -2 .

(ii) A temperature of -1°C is _____ a temperature of 6°C .

(iii) Floor 3 is _____ floor -1 .

B2 All four rules, and all four signs

Work these out. You may draw a small number line beside any of them.

(a) $-7 + 3 =$ _____

(e) $-8 \times 3 =$ _____

(i) $-20 + (-14) =$ _____

(b) $6 + (-11) =$ _____

(f) $-6 \times (-7) =$ _____

(j) $12 - (-12) =$ _____

(c) $-4 - 9 =$ _____

(g) $45 \div (-9) =$ _____

(k) $-4 \times (-5) \times (-2) =$ _____

(d) $-5 - (-12) =$ _____

(h) $-72 \div (-8) =$ _____

(l) $-30 \div 6 \times (-2) =$ _____

B3 Say it, then write it

Word help – these words decide the order of your calculation

subtract 5 from 8 you **start** at 8: $8 - 5$. The English gives the numbers in the **opposite order** to the calculation.

take 6 away from 2 the same trap: $2 - 6$.

7 less than 3 again the same trap: $3 - 7$.

owe / a debt money you must pay back. It makes your total **negative**.

pay back to give owed money **back**, so the debt gets **smaller**.

Turn each English sentence into a calculation, then work it out. **Write the calculation first.**

1. Subtract 11 from 5.

Calculation: _____ Answer: _____

2. Take 8 away from -3 .

Calculation: _____ Answer: _____

3. What is 9 less than 2?

Calculation: _____ Answer: _____

4. Mr Bowen owes 45 dollars. He pays back 20 dollars.

Calculation: _____ Answer: _____

B4 Difference

Word help – difference

The **difference** between two numbers is the **gap** between them, so it is never negative – and **the difference between a and b** is **not** the same instruction as **a minus b** . Question 2 is about exactly that.

1. What is the difference between 12 and -5 ? _____

2. (a) The difference between -7 and 4 is _____ (b) $-7 - 4 =$ _____

(c) These two answers are not the same. Explain why.

Section C · Mathematics 1.3–1.6 – Factors, Multiples, Tests and Roots

C1 Build the HCF and the LCM from the lists

One list stops, the other never does

Factors divide into the number. That list **stops**, so we ask for the **highest** common one – the **HCF**.

Multiples are its times table. That list **never stops**, so we ask for the **lowest** common one – the **LCM**.

- (a) All the factors of 48: _____

- (b) All the factors of 60: _____

- (c) The common factors: _____ so the **HCF** is _____

- (d) The first six multiples of 8: _____

The first six multiples of 12: _____

So the **LCM** of 8 and 12 is _____

(e) A student says the LCM of 8 and 12 is 96, because $8 \times 12 = 96$. Is 96 a common multiple of 8 and 12? Is it the **lowest** one? Explain.

C2 Which one does the question want?

Word help – the words that tell you which tool to reach for

together again · **at the same time** · **how often** – something **repeating**. You want the **LCM**.

share into equal groups · **as large as possible** · **none left over** – something being **split up**. You want the **HCF**.

Tick **HCF** or **LCM** for each one, then work out the answer.

The question	HCF	LCM	Answer
One bell rings every 6 minutes, another every 10. They ring together now. In how many minutes will they ring together again?			
Mr Bowen has 24 pens and 36 pencils. He makes identical packs, using everything, with none left over. What is the greatest number of packs?			
Ribbons 45 cm and 60 cm long are cut into equal pieces, as long as possible, with none wasted. How long is each piece?			
A red bus leaves every 12 minutes and a blue bus every 18 minutes. They leave together at 7 am. When do they next leave together?			

C3 Tests for divisibility

(a) Put a tick (✓) where the number divides exactly. Use the tests – do **not** do the divisions.

Number	2	3	4	5	6	9
3942						
5280						
7128						
2475						

(b) **Three sentences, one fact.** We know that $48 \div 6 = 8$ exactly. Fill in the three ways of saying so.

- (i) 6 is a _____ of 48. (ii) 48 is a _____ of 6.
(iii) 48 is _____ by 6.

C4 Squares, cubes and roots

Rooting is squaring run backwards

$9^2 = 81$, so $\sqrt{81} = 9$. $4^3 = 64$, so $\sqrt[3]{64} = 4$. And a square number can never end in **2, 3, 7 or 8**.

- (a) $7^2 =$ _____ (d) $10^3 =$ _____ (g) $\sqrt[3]{27} =$ _____
(b) $11^2 =$ _____ (e) $\sqrt{81} =$ _____ (h) $\sqrt[3]{125} =$ _____
(c) $4^3 =$ _____ (f) $\sqrt{144} =$ _____ (i) $6^2 + 3^3 =$ _____

(j) Two of these **cannot** be square numbers. Ring them, and say how you know without doing any working.

2500 4832 6400 1007

(k) **Consecutive** numbers follow each other, like 5, 6, 7. The consecutive pair whose squares are 64 and 81 is _____

Section D · Mathematics — Word Problems

Every question here is one of the six maths lessons wearing a story. Before you calculate, ask yourself **which lesson this is** — then write the calculation down before the answer.

1. **Sa Pa.** At 5 am the temperature was -4°C . By 1 pm it had risen by 13 degrees. By 9 pm it had fallen by 9 degrees. **(a)** What was the temperature at 9 pm? **(b)** What is the difference between the warmest and the coldest temperature of the day?

2. **The lighthouses.** One lighthouse flashes every 8 seconds and another every 12 seconds. They flash together at exactly 9:00:00. At what time do they next flash together?

3. **The pens.** Mr Bowen has 54 red pens and 36 blue pens. He makes identical packs, using every pen, with none left over, and he wants **as many packs as possible**. How many packs, and how many of each colour goes in one pack?

4. **The patio.** **(a)** A square patio is made of 169 square tiles. How many tiles are there along one side? **(b)** A large cube is built from 216 small cubes. How many small cubes are there along one edge?

5. **The biscuits.** Mr Bowen buys a bag of 60 round biscuits and a bag of 48 square biscuits. He puts them on plates so that every plate has the same number of round biscuits and the same number of square biscuits, with none left over, using **as many plates as possible**. **(a)** How many plates are there? **(b)** How many biscuits are on one plate? **(c)** Mr Bowen then eats one whole plate. How many biscuits are left?

D2 Now you write the question**The rules for both**

A **real** situation – money, temperature, floors, buses, packing. Full sentences, ending in a **question**. At least one of the **signal words** from this packet. And your own question must actually come out at the answer given.

Write a word problem for this calculation.

$$-7 + 12 = 5$$

Answer Key – Teacher Copy

A1 Match the word to its meaning

1-D 2-F 3-H 4-A 5-C 6-G 7-E 8-B

The pair to watch is 2 and 3. A student who swaps *tissue* and *organ* has the ladder upside down and will get A2 wrong as well – mark the two together and re-teach once, rather than twice.

A2 Climb the ladder

A red blood cell	cell	tissue (blood)
Muscle	tissue	organ (the heart, for example)
The stomach	organ	organ system (the digestive system)
The digestive system	organ system	organism
Mr Bowen	organism	nothing – this is the top rung

The last row is the question, and it is worth discussing rather than marking. Most of the class will invent a rung – “a family”, “a school” – because the table has a box and a box wants filling. That instinct is the thing to name out loud: **an empty box can be a correct answer**. Accept anything saying there is no level above an organism.

A3 Name the cell

(i) red blood cell (ii) neurone / nerve cell (iii) ciliated cell (iv) root hair cell (v) palisade cell

(b) For example: “A root hair cell is adapted to taking in water because it has a long thin part that reaches into the soil.”

(c) A root hair cell is underground, where there is no light, so chloroplasts would be no use to it. A palisade cell is near the top of the leaf where the light is, and its job is to make food.

In (b), mark the *frame*, not the biology. The sentence must contain both *adapted to* + a job and *because it has* + a feature. A student who writes “it has a long thin part” has the science right and the sentence wrong, and it is the sentence being taught here.

(c) separates recall from understanding. “Because it does not need them” is not yet an answer – push for *no light underground*.

A4 Answer in full sentences

A plant cell. The straight stiff edge is the cell wall, and the large space in the middle is the sap vacuole; an animal cell has neither.

This question wants the *evidence*, not the verdict. “Plant” on its own is half an answer, and the missing half is the half being taught – push for both structures to be named, in sentences.

A5 Label the animal cell

1 – cell membrane 2 – cytoplasm 3 – nucleus 4 – mitochondrion

Now think: any two of – cell wall, chloroplasts, sap vacuole.

Expect “cell wall” for 1. The diagram has one outline on purpose and an animal cell has no wall, so this is a real error and not a slip: the outside of an animal cell is the membrane. That is the whole reason this packet labels an animal cell rather than repeating the plant cell of Homework 1.

B1 Higher, lower, warmer, colder

(i) lower than (ii) colder than (iii) higher than.

The way in, and it should take a minute. If (i) goes wrong, the number line is still not automatic and everything below it will be arithmetic done by guesswork.

B2 All four rules, and all four signs

(a) -4 (b) -5 (c) -13 (d) 7 (e) -24 (f) 42

(g) -5 (h) 9 (i) -34 (j) 24 (k) -40 (l) 10

(i) is the whole point of putting the two rule sets side by side. $-20 + (-14)$ is an **addition**, so the two negatives do **not** make a positive; a student who writes 34 has over-applied the multiplying rule, and that error deserves more class time than anything else on this page.

(k) has three negatives and comes out negative — they cancel in pairs, and one is left over. (l) must be worked left to right; a student who does $6 \times (-2)$ first gets 2.5 and should be shown why.

B3 Say it, then write it

1. $5 - 11 = -6$

2. $-3 - 8 = -11$

3. $2 - 9 = -7$

4. $-45 + 20 = -25$

1–3 are the reversed-order trap. If a student writes $11 - 5$, $8 - (-3)$ or $9 - 2$, that is an **English** error, not an arithmetic one — do not correct the sum, make them re-read the sentence and say which number they start at.

4 is the one to check: paying back a debt makes the total go *up*, and a student who writes $-45 - 20$ has read “pays” as “spends”.

B4 Difference

1. 17

2. (a) 11 (b) -11 (c) A difference is the gap between the two numbers, so it is never negative. $-7 - 4$ is an instruction to start at -7 and move 4 further left.

Part (c) is the question. Accept any answer saying the first is a *distance* and the second is a *move*. This is Homework 1’s trap with new numbers, and if the class still splits on it, it is worth ten minutes on the board rather than a tick and a cross.

C1 Build the HCF and the LCM from the lists

(a) $1, 2, 3, 4, 6, 8, 12, 16, 24, 48$

(b) $1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60$

(c) $1, 2, 3, 4, 6, 12$; HCF = **12**

(d) $8, 16, 24, 32, 40, 48$ and $12, 24, 36, 48, 60, 72$; LCM = **24**

(e) 96 is a common multiple — 8×12 always is. It is not the **lowest**, because 24 is smaller and is also in both lists.

The common miss in (a) and (b) is a method problem. Students who hunt in pairs (1×48 , 2×24 , 3×16 , 4×12 , 6×8) get all ten; students who count upwards in ones stop early and lose 16 and 24 . Point at the method, not the missing number.

(e) is worded so that “yes” and “no” are both needed. A student who answers only “no” has not read both questions.

C2 Which one does the question want?

Two bells, 6 and 10 minutes	LCM	30 minutes
24 pens and 36 pencils	HCF	12 packs (2 pens and 3 pencils in each)
Ribbons 45 cm and 60 cm	HCF	15 cm each
Buses every 12 and 18 minutes	LCM	36 minutes, so 7:36 am

Mark the ticks before the answers. A right tick with wrong arithmetic is a good piece of work; a wrong tick with tidy arithmetic is the error this whole packet aims at. The give-away is size: a student who answers 2 minutes for the bells, or 180 cm for the ribbon, has reached for the wrong tool – and asking “could a piece really be longer than the ribbon?” fixes it faster than re-teaching the method. The bus question wants a *time* back – 7:36 am, not 36 – because it asked “when”. That is the same reading habit as everything in Section D.

C3 Tests for divisibility

3942	2, 3, 6, 9	(digit sum 18; last two digits 42, not a 4)
(a) 5280	2, 3, 4, 5, 6	(digit sum 15, so 3 but not 9)
7128	2, 3, 4, 6, 9	(digit sum 18; 28 divides by 4)
2475	3, 5, 9	(odd, so no 2 – and therefore no 6)

(b) (i) **factor** (ii) **multiple** (iii) **divisible**

In (a), the row to check is 2475: a student who ticks 6 has not noticed that 6 needs *both* 2 and 3, and 2475 is odd.

(b) is a language question wearing arithmetic, and it is the one most likely to be left blank. Three sentences describe one fact, and the class does not yet believe that.

C4 Squares, cubes and roots

- (a) 49 (b) 121 (c) 64 (d) 1000 (e) 9 (f) 12 (g) 3 (h) 5 (i) $36 + 27 = 63$
 (j) **4832** and **1007** – a square number never ends in 2, 3, 7 or 8. ($2500 = 50^2$ and $6400 = 80^2$.)
 (k) **8** and **9**.
 (i) is the one that goes wrong, and not because of the arithmetic: students read 3^3 as 3×3 and answer 45. Squaring and cubing look almost identical on the page, which is exactly why the two lists were copied out separately.
 (j) rewards *not* calculating, so praise the student who answered it in five seconds. *Consecutive* in (k) is the word the textbook uses without ever explaining it.

D1 Word problems

- (a) $-4 + 13 - 9 = 0^\circ\text{C}$. (b) Warmest 9°C at 1 pm, coldest -4°C at 5 am: $9 - (-4) = 13$ degrees.
- LCM of 8 and 12 is 24, so **9:00:24**.
- HCF of 54 and 36 is 18. **18 packs**, each with **3 red** and **2 blue**.
- (a) $\sqrt{169} = 13$ tiles. (b) $\sqrt[3]{216} = 6$ cubes.
- HCF of 60 and 48 is 12. (a) **12 plates**. (b) $5 + 4 = 9$ biscuits. (c) $108 - 9 = 99$ left.

The diagnosis is almost never the arithmetic.

1(a) comes out at 0°C , and that is the question: a student who wrote the calculation down will trust it, and a student who did not will decide zero “looks wrong” and change it. This is the question to make the write-it-first argument with.

2 must come back as a *time*; “24” is an unfinished answer.

3 asks two things and half the class will answer one.

5 is three steps and the last one is a subtraction, not a division — a student who answers 12 or 9 stopped reading. Part (b) needs both halves, $60 \div 12 = 5$ and $48 \div 12 = 4$; answering “5” is the same stopped-reading error one step earlier.

D2 Now you write the question

No single right answer. Check the conditions in this order:

1. **Does the story start at -7 ?** It must *begin* there, not arrive there — the same “which number comes first” problem as B3, and the one most of them will get wrong.
2. **Is there a signal word, and does it match?** The $+12$ needs an *up* word: rises, deposits, climbs, pays back.
3. **Full sentences, ending in a question?**
4. **Does their own question actually come out at 5?**

One that works: “The temperature in Sa Pa was -7°C at midnight. By noon it had risen by 12 degrees. What was the temperature at noon?”

A question that simply ends “what is $-7 + 12$?” has not done the task — it is the calculation copied out in words, and that is worth naming out loud rather than marking wrong. This is the single best question on the packet for finding out who is reading and who is pattern-matching, so it is worth reading four or five of them to the class.